P-S-579A

**P-S-579a**
**March 6, 1963**

**SUPERSEDING**
**Int. Fed. Spec. P-S-00579 (Army-QMC)**
**March 20, 1962,**
**Fed. Spec. P-S-580a**
**March 20, 1959, and**
**Fed. Spec. P-S-596c**
**May 26, 1952**

**FEDERAL SPECIFICATION**

# SOAP, LAUNDRY, CHIP AND POWDERED

*This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.*

## 1. SCOPE AND CLASSIFICATION

**1.1 Scope.** This specification covers two types of laundry soaps (see 6.1).

**1.2 Classification.**

**1.2.1** *Grade and types.* The soaps covered by this specification shall be of one grade and shall be of the following types, as specified (see 6.2):

Type I—Chip or flake.
Type II—Powdered or granular.

## 2. APPLICABLE SPECIFICATIONS AND STANDARDS

**2.1** The following specifications and standards, of the issues in effect on date of invitations for bids, form a part of this specification to the extent specified herein.

*Federal Specifications*:

L-P-378—Plastic Film (Polyethylene Thin Gage).
RR-S-366—Sieves, Standard for Testing Purposes.
UU-S-48—Sacks, Shipping, Paper.
PPP-B-35—Bags, Textile, Shipping, Burlap.
PPP-B-636—Box, Fiberboard.
PPP-D-723—Drums, Fiber.

*Federal Standards*:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 536—Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing.

Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

**FSC 7930**



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:18Z
Check the source to verify that this is the current version before use.

*Military Standards*:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.
MIL-STD-147—Palletized Unit Loads.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer).

## 3. REQUIREMENTS

**3.1 Preproduction sample.** When specified in the contract or order (see 6.2) and before production begins, complete test results showing that the product conforms to the requirements in this specification and a 5-pound sample of the applicable end item shall be submitted to the contracting officer for approval. Approval of the preproduction sample authorizes production. Such authorization does not relieve the contractor of the responsibility for compliance with all provisions of this specification.

**3.2 Composition.** The composition of types I and II soaps shall conform to the requirements specified in table I, when tested as specified in table II (see 6.3).

**3.2.1** *Computation of volatile matter.* The percentage of moisture and volatile matter at 105°C. shall be computed and reported by the testing laboratory on the soaps as received. The percentage of all other constituents shall be calculated and reported on a moisture and volatile matter content of 10 percent for type I and 6 percent for type II. (For basis of purchase see 6.3.) To adjust the percentage of the other constituents, where applicable, for samples having a moisture and volatile matter content at 105°C. of less than 10 percent for type I and 6 percent for type II, multiply the percentage of constituents requirement, other than moisture and volatile matter, by a factor (F) as follows:

For Type I:

$$F = \frac{100 - 10}{100 - \text{percent moisture and volatile matter (as received)}}$$

For Type II:

$$F = \frac{100 - 6}{100 - \text{percent moisture and volatile matter (as received)}}$$

**3.3 Color.** The soaps shall have a uniform light color. Unless otherwise specified (see 6.2), the color of a freshly sampled soap shall be not darker than color number 30266 of Fed. Std. No. 595 (see 6.4).

TABLE I.—*Composition*

| Characteristic | Type I | | Type II | |
|---|---|---|---|---|
| | Minimum | Maximum | Minimum | Maximum |
| Moisture and volatile matter at 105°C., percent | — | 10.0 | — | 6.0 |
| Sum of free alkali or free acid, total matter insoluble in alcohol, and sodium chloride, percent | — | 6.0 | — | 6.0 |
| Free alkali, calculated as sodium hydroxide, (NaOH), percent | — | 0.2 | — | 0.2 |
| Free acid, calculated as oleic acid, percent | — | 0.2 | — | 0.2 |
| Matter insoluble in water, percent | — | 1.0 | — | 1.0 |
| Anhydrous soap, percent | 85.0 | — | 89.0 | — |
| Titer of the mixed fatty acids prepared from the soap, °C. | 39.0 | — | 39.0 | — |
| Rosin | — | 0 | — | 0 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:18Z
Check the source to verify that this is the current version before use.

**3.4 Odor.** The soaps shall be free of an objectionable odor as received and in a solution of the soap in water at 130°-140°F. when tested as specified in 4.4.2. An objectionable odor, in this specification, is defined as a rancid, fishy, or horsey odor.

**3.5 Form.**

**3.5.1** *Type I.* The soap shall be in chip or flake form and no particle shall exceed 3 grams in weight when tested as specified in 4.4.3.

**3.5.2** *Type II.* The soap shall be powdered or granular form and a maximum residue of 1.5 percent shall be retained on a number 12 sieve when tested as specified in table II.

**3.6 Workmanship.** The finished product shall be clean, uniform, free flowing and shall not lump or cake.

**3.7 Quality.** The quality of the finished product shall conform to the levels established herein. The occurrence of defects shall not exceed the number permitted under the acceptable quality levels (AQLs) specified in section 4 for any examinations and tests of materials, components and end items, as applicable.

## 4. SAMPLING, INSPECTION, AND TEST PROCEDURES

**4.1 Inspection responsibility.** The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

**4.1.1** *Sampling for inspection and tests.* Sampling for inspection and tests shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, a lot shall consist of all end items of the same type and size, made from the same components, and submitted for inspection and test at one time.

**4.2 Inspection of materials and components.** In accordance with 4.1, the supplier is responsible for insuring that all the materials and components used in the end item were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification. In the event of conflict, this specification shall govern.

**4.3 Inspection of the end item.**

**4.3.1** *Examination of the end item.* The end item shall be examined in accordance with the classification of defects set forth in the applicable sub-paragraphs at the inspection levels and acceptable quality levels (AQLs) in 4.3.6. Random samples shall be drawn from each lot of end items offered for inspection. The lot size for purposes of determining the sample size, in accordance with MIL-STD-105 shall be expressed in units of filled primary containers of the same capacity, consisting of soap of the specified type, manufactured in one batch using the same materials, and offered for the examination in 4.3.2, 4.3.3, and 4.3.4, and in units of shipping containers for examination under 4.3.5.

**4.3.2** *Examination of the primary container for visual defects in construction and markings.* The sample unit for this examination shall be one filled primary container.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:18Z
Check the source to verify that this is the current version before use.

| Examine | Defect |
|---|---|
| Construction | Not type specified.<br>Any seepage of contents.<br>Any split, break, dent, hole, tear, puncture, or other defect which may affect serviceability.<br>Seams or closures not as specified. |
| Markings | Omitted; illegible; incomplete; smeared; incorrect; not as specified. |

4.3.3 *Examination of the soap product for defects in workmanship, color and odor.* The sample unit for this examination shall be the contents of one filled primary container.

| Examine | Defect |
|---|---|
| Workmanship | Not type specified (chip or powdered).<br>Not clean; not uniform; any foreign matter.<br>Not free flowing.<br>Lumping or caking. |
| Color | Not uniform.<br>Too dark; darker than color chip number 30266, unless otherwise specified. |
| Odor | Any objectionable odor (rancid, fishy, or horsey odor). |

4.3.4 *Examination for defects in net contents.* The sample unit for this examination shall be one filled primary container. The average net contents per container shall be not less than the specified or indicated quantity.

4.3.5 *Examination of preparation for delivery.* An examination shall be made to determine that packaging (as applicable), packing, count, weight, and markings comply with the requirements of section 5 of this specification. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Closed shipping containers, fully prepared for delivery shall be examined for closure defects. This examination is not applicable when the primary container is the shipping container.

| Examine | Defect |
|---|---|
| Packaging (as applicable) | Not in accordance with contract requirements.<br>Unit packages not packaged as specified; packaging material not as specified. |
| Packing | Not in accordance with contract requirements.<br>Container not specified; closures not accomplished by specified or required methods or materials.<br>Any nonconforming component; component missing, damaged, or otherwise defective affecting serviceability.<br>Inadequate application of components: Such as incomplete closure of case liners, container flaps; loose or inadequate sealing, strapping, or stapling.<br>Bulged or distorted containers.<br>Not palletized when specified. |
| Count | Number of packages per shipping container less than the specified or indicated quantity. |
| Weight | Gross weight exceeds contract requirements. |
| Markings, exterior and interior as applicable | Omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements. |

4.3.6 *Inspection levels and acceptable quality levels (AQLs) for examinations.* The inspection levels, for determining the sample size, and the acceptable quality levels (AQLs) expressed in defects per one hundred units shall be as follows:

| Examination paragraph | Inspection levels | AQLs |
|---|---|---|
| 4.3.2 ..... | I | 2.5 |
| 4.3.3 ..... | L-4 | 4.0 |
| 4.3.4 ..... | L-3 | — |
| 4.3.5 ..... | L-4 | 4.0 |

4.3.7 *Testing of the soap product.* The soap shall be tested for the characteristics



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:18Z
Check the source to verify that this is the current version before use.

TABLE II.—*Instructions for testing*

| Characteristic | Specification reference | | Number determinations on composite sample | Results reported at | |
|---|---|---|---|---|---|
| | Requirement | Test method[1] | | Pass or fail[2] | Numerically to nearest[3] |
| Moisture and volatile matter at 105°C. | Table I | 201.1 | Average of 2 | | 0.1 percent |
| Sum of free alkali or free acid, total matter insoluble in alcohol, and sodium chloride (NaCl) | Table I | 302.3<br>301.3<br>501 | Average of 2 | | 0.1 percent |
| Free alkali, calculated as sodium hydroxide (NaOH) | Table I | 302.3 | Average of 2 | | 0.1 percent |
| Free acid, calculated as oleic acid | Table I | 302.3 | Average of 2 | | 0.1 percent |
| Matter insoluble in water | Table I | 303.1 | Average of 2 | | 0.1 percent |
| Anhydrous soap | Table I | 401.3 | Average of 2 | | 0.1 percent |
| Titer of the mixed fatty acids prepared from the soap | Table I | 901 | Average of 2 | | 0.1°C. |
| Rosin | Table I | 701.3; Test No. 2 | 1 | X | |
| Odor | 3.4 | 4.4.2 | 1 | X | |
| Particle weight (type I only) | 3.5.1 | 4.4.3 | 1 | X | |
| Residue retained on a number 12 sieve (type II only) | 3.5.2 | 2101 | Average of 2 | | 0.1 percent |

[1] All tests except odor and particle weight are performed in accordance with Fed. Test Method Std. No. 536.

[2] If failure is indicated, report description of any failures.

[3] Test reports shall include all values on which results are based.

in table II for each lot presented for examination. The sample unit, for purposes of testing, shall be one pound composite sample selected in accordance with 4.3.8. When duplicate determinations are specified on the composite sample, the average of the two (2) determinations shall comply with the requirements.

4.3.8 *Composite sample.* When four or more containers comprise the shipping lot, a sample of not less than one pound shall be taken from each of not less than four separate shipping containers, randomly selected throughout the lot, presented for examination. When less than four shipping containers comprise the lot, the total samples shall, in all cases, consist of not less than three sample portions taken at random from the containers or container. The samples shall be placed in a receptacle of suitable size. After mixing thoroughly, a one pound com-



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:18Z
Check the source to verify that this is the current version before use.

posite sample shall be drawn for testing purposes. The composite sample shall be placed in an air-tight container, sealed, and marked with the identifying lot number and date.

**4.4 Test procedures.**

**4.4.1** *Composition.* The composition (table I) shall be determined according to the methods specified in Fed. Test Method Std. No. 536 as indicated in table II.

**4.4.2** *Test for odor of soap solution.* A one-ounce-per-gallon of distilled or demineralized water soap solution shall be prepared and heated to 130°-140°F. Determine the odor of the soap solution.

**4.4.3** *Particle weight (type I).* Ten of the largest observed particles shall be selected from the composite sample. Each particle shall be weighed separately. The weight of each particle shall not exceed three grams.

## 5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurements.)

**5.1 Packaging.** Packaging shall be level A or C, as specified (see 6.2).

**5.1.1** *Levels A and B.* Soap shall be packaged in quantities and containers listed in table III, as specified (see 6.2).

**5.1.1.1** *Paperboard box.* Paperboard box shall conform to those used in the supplier's normal commercial practice. The box shall be adequately protected against sifting of contents. Liners or wraps, when used, shall be of paper or polyethylene and shall not affect or be adversely affected by the soap.

**5.1.1.2** *Fiber drum.* Fiber drum shall conform to type I or III, grade A of PPP-D-723 (see 5.2.1.2). Each drum shall be lined with a minimum 0.002-inch thick polyethylene bag fabricated from material conforming to type I, grade A, finish 1 of L-P-378. All seams and closures shall be effected by heat-sealing and shall conform to the heat seal strength requirements of L-P-378.

**5.1.1.3** *Bag.* Bag shall be a textile shipping bag or paper shipping sack conforming to bag No. C51 or C56, with liner C, D, or E of PPP-B-35, or construction number 10L/W, type optional of UU-S-48, respectively (see 5.2.1.2).

**5.1.2** *Level C.* Soap shall be packaged in accordance with the industry's practice.

**5.2 Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

**5.2.1** *Level A.*

**5.2.1.1** *Paperboard boxes.* Paperboard boxes containing only one type of soap shall be packed in the quantity specified in table III in a snug-fitting fiberboard shipping container conforming to class 2, V3c or V3s,

TABLE III.—*Packaging and packing of soap*

| Quantity of soap per package | Container for package | Quantity of packages per pack | Shipping container |
|---|---|---|---|
| 1 pound | Paperboard box (see 5.1.1.1) | 48 | Fiberboard |
| 1½ pounds | Paperboard box (see 5.1.1.1) | 36 | Fiberboard |
| 5 pounds | Paperboard box (see 5.1.1.1) | 12 | Fiberboard |
| 100 pounds | Fiber drum or bag, as specified (see 5.1.1.2, 5.1.1.3, and 6.2) | — | — |
| 200 pounds | Fiber drum (see 5.1.1.2) | — | — |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:18Z
Check the source to verify that this is the current version before use.

style RSC of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape or a case liner, and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification.

**5.2.1.2** *Fiber drums and bags.* Soap, packaged in fiber drums or bags as specified in 5.1.1.2 or 5.1.1.3 (except that paper shipping sacks and type I fiber drums shall not be used) shall require no overpacking. When specified, fiber drums and textile shipping bags shall be palletized in accordance with MIL-STD-147 (see 6.2).

**5.2.2** *Level B.*

**5.2.2.1** *Paperboard boxes.* Paperboard boxes containing only one type of soap shall be packed in the quantity specified in table III in a snug-fitting fiberboard shipping container conforming to type I or II, class 1, style RSC of PPP-B-636. All flaps of the fiberboard shipping container shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps or by a combination of metal-stitching the bottom flaps and sealing the top flaps with adhesive.

**5.2.2.2** *Fiber drums and bags.* Soap packaged in fiber drums or bags as specified in 5.1.1.2 or 5.1.1.3 shall require no overpacking. When specified, fiber drums, textile shipping bags, and paper shipping sacks shall be palletized in accordance with MIL-STD-147 (see 6.2).

**5.2.3** *Level C.* Soap, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

**5.3 Marking.**

**5.3.1** *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

**5.3.2** *Military requirements.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129. Each shipping container shall be marked with the date of manufacture of the soap.

## 6. NOTES

**6.1 Intended use.** This specification covers neutral laundry soaps substantially free from uncombined alkali or fatty matter. The soaps are intended for laundering and general cleaning purposes with soft water, where the presence of alkaline salts is not desirable. The soaps are suitable for use in high temperature (140°—160°F.) laundering of cotton fabrics. It may also be used with added alkali in the laundering process.

**6.2 Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type required (see 1.2).

(*c*) Whether or not a preproduction sample is required (see 3.1).

(*d*) Shade of color, when required (see 3.3).

(*e*) Inspection responsibility, methods, and tests (see section 4).

(*f*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(*g*) Quantity of soap per package (see 5.1.1).



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:18Z
Check the source to verify that this is the current version before use.

(*h*) Whether the 100-pound quantity is to be packaged in a fiber drum or bag (see table III).

(*i*) If drums, bags, or sacks are to be palletized (see 5.2.1.2 and 5.2.2.2).

**6.3 Basis purchase.** Unless otherwise specified, sampling of soap shall be performed at manufacturers plant.

6.3.1 *Type I.*

6.3.1.1 The soap should be purchased by net weight provided the matter volatile at 105°C. is neither above nor below 8 percent.

6.3.1.2 Deliveries containing more than 10 percent of matter volatile at 105°C. should be rejected without further test.

6.3.1.3 On deliveries containing 10 percent or less of matter volatile at 105°C., payment should be made on a basis of a product containing 8 percent moisture. The net weight of the material to be paid for on an 8 percent moisture and matter volatile basis is calculated as follows:

$$W = \frac{R \times (100\text{-}L)}{92}$$

Where:

W = Net weight of material to be paid for on 8 percent moisture and matter volatile basis.
R = Net weight of material as received.
L = Percentage of loss at 105°C.

For example: for 9 percent moisture and volatile matter where:

$$W = \frac{1 \times (100\text{-}9)}{92} = 0.989 \text{ pound}$$

Cost of soap = number of pounds of soap as received × 0.989 × price per pound.

Likewise, for soap containing less than 8 percent moisture, such as 6 percent.

$$W = \frac{1 \times (100\text{-}6)}{92} = 1.022 \text{ pounds}$$

Cost of soap = number of pounds of soap × 1.022 × price per pound.

6.3.2 *Type II.*

6.3.2.1 The material should be purchased by net weight, provided the matter volatile at 105°C. is neither above nor below 4 percent.

6.3.2.2 Deliveries containing more than 6 percent of matter volatile at 105°C. should be rejected without further test.

6.3.2.3 On deliveries containing 6 percent or less of matter volatile at 105°C., payment should be made on a basis of a product containing 4 percent moisture; that is, 0.96 pound of nonvolatile matter should be considered 1 pound of soap. The net weight of the material to be paid for on a 4 percent moisture and matter volatile basis is calculated as follows:

$$W = \frac{R \times (100\text{-}L)}{96}$$

Where:

W = Net weight of material to be paid for on 4 percent moisture and matter volatile basis.
R = Net weight of material as received.
L = Percentage of loss at 105°C.

For example: For 5 percent moisture and volatile matter where:

$$W = \frac{1 \times (100\text{-}5)}{96} = 0.990 \text{ pound}$$

Cost of soap = number of pounds of soap as received × 0.990 × price per pound.

Likewise, for soap containing less than 4 percent moisture, such as 3 percent.

$$W = \frac{1 \times (100\text{-}3)}{96} = 1.010 \text{ pounds}$$

Cost of soap = number of pounds of soap × 1.010 × price per pound.

6.4 Color chips are available from all General Services Administration Business Service Centers.

**6.5 Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:18Z
Check the source to verify that this is the current version before use.

*Rail*:

Soap, not otherwise indexed by name other than liquid.

Carload minimum weight 36,000 pounds.

*Motor*:

Soap, not otherwise indexed, other than liquid.

Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

**Notice.** When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

**MILITARY CUSTODIANS:**

**Army—GL**
**Navy—Sh**
**Air Force—MOA**

**OTHER INTEREST:**

**Army—M**



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:18Z
Check the source to verify that this is the current version before use.